

Sys_3_4: An Enterprise Context-Aware and Dual-Engine Legal AI Assistant for Vietnamese Public Administration

A Production-Ready Architecture Combining Grounded Information Retrieval, Multi-Turn
State Management, and Local SLMs

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Abstract

Public administration in Vietnam is undergoing rapid national digital transformation under governmental directives such as Project 06. However, ordinary citizens encounter substantial barriers when navigating grassroots-level administrative procedures (commune and ward levels). These difficulties arise from complex statutory language, multi-tiered regulatory decrees, and colloquial citizen queries characterized by shorthand, unaccented typing, multi-intent requests, and personal demographic conditions. General-purpose Large Language Models (LLMs) frequently exhibit hallucinations regarding administrative fees and statutory processing deadlines, succumb to prompt injection attacks, and lose context across extended multi-turn dialogues.

This paper presents **Sys_3_4**, an enterprise-grade, domain-adaptable legal artificial intelligence assistant that unifies a deterministic, zero-hallucination **Strict Engine (System 3)** with a conversational, domain-flexible **Friendly Engine (System 4)**. **Sys_3_4** resolves multi-turn conversational degradation via: (1) A deterministic dialogue state machine equipped with *Clarify State Isolation* that eliminates phantom procedure contamination during disambiguation turns; (2) A colloquial demographic condition extractor tailored to Vietnamese phrasing; (3) A vector-based Semantic Cache (GPTCache) on Redis; (4) A dynamic query complexity router (RouteLLM); and (5) A paged memory serving engine using PagedAttention (vLLM).

We report comprehensive, unembellished empirical evaluations across both controlled developmental suites and unseen blind test sets. On the official blind evaluation suite (**HOLDOUT-4**, 90 items / 106 dialogue turns), **Sys_3_4** attains a **75.3% Top-1 retrieval accuracy** (58/77) and **82.1% behavioral accuracy** (87/106), compared to only 10.6% Top-1 and 31.9% behavioral accuracy in the initial system baseline. On the controlled developmental benchmark (DEV, 209 cases), it achieves **96.3% Top-1** and **97.6% behavioral accuracy** with **0.0% numerical hallucination**. On real-world municipal cases (10 team test queries), the system scores **7/10 PASS**, with failures transparently attributed to portal snapshot data omissions. Operating in pure deterministic rule mode, **Sys_3_4** delivers a median latency of **20–33 ms** (p95 of 49–80 ms) on consumer edge hardware, demonstrating that deterministic RAG with disciplined context isolation outperforms over-parameterized LLM planning for public service delivery.

Keywords: Sys_3_4, Legal AI Assistant, Vietnamese Public Administration, Multi-Turn Context Man-

agement, Retrieval-Augmented Generation (RAG), Semantic Cache, Deterministic Grounding, Empirical Evaluation.

1 Introduction

Grassroots public administration at the commune, ward, and township levels constitutes the primary frontline interface between citizens and state governance in Vietnam. It encompasses essential civil procedures such as vital registration (birth registration, marriage registration, death registration, guardianship), household business registration, social welfare assistance, social insurance, and residency declarations. While the National Public Service Portal (`dichvucong.gov.vn`) provides digitized access to statutory procedures, ordinary citizens continue to experience significant hurdles:

1. **Discrepancy Between Statutory Language and Colloquial Inquiries:** Official administrative announcements utilize strictly codified legal vocabulary, whereas citizens express inquiries in informal dialects, teencode shorthand, unaccented text, or vernacular idioms (e.g., informal phrasing for marriage, childbirth, or temporary lodging).
2. **Multi-Turn Dialogue Dependency and Demographic Specifics:** Inquiries rarely manifest as self-contained prompts. Citizens typically initiate conversation with broad queries (“I want to get married”), followed by fragmentary quantitative inquiries (“How long does it take and how much does it cost?”), and subsequently introduce personal qualifiers (“What additional documents are required if one partner was previously divorced?”).
3. **Stringent Zero-Tolerance for Hallucination in Public Governance:** In administrative law, inaccuracies carry direct legal and financial repercussions. An AI assistant hallucinating fee exemptions, fabricating expediting deadlines, or directing citizens to improper jurisdictional bodies severely disrupts civic compliance and damages institutional trust.
4. **On-Premise Infrastructure Constraints and Data Sovereignty:** Local government agencies operate within air-gapped or compute-constrained server environments, precluding reliance on high-latency, costly commercial cloud APIs that risk leaking citizen Personally Identifiable Information (PII).

To address these compounding challenges, this work introduces **Sys.3.4**, an enterprise-scale architecture that integrates the deterministic verification core of System 3 with the multi-domain conversational layer of System 4.

The primary scientific and engineering contributions of Sys.3.4 include:

- **Dual-Engine Architecture:** Decoupling the pipeline into a *Strict Mode* (governed by statutory grounded citations, deterministic BM25/FTS5 indexes, and verifiable safety gates) and a *Friendly Mode* (driven by local SLMs, conversational personas, and Model Context Protocol tool integrations).
- **Robust Multi-Turn Context State Engine:** Developing a state tracking mechanism that incorporates *Clarify State Isolation* to eliminate “phantom procedure” memory pollution during disambiguation turns, alongside dynamic field inheritance on dialogue return paths.

- **Production-Grade Serving Infrastructure:** Integrating semantic vector caching (GPTCache), dynamic fast/slow path routing (RouteLLM), continuous paged KV-cache serving (vLLM), and layout-preserving document ingestion (Nougat).
- **Transparent, Unembellished Empirical Evaluation:** Benchmarking on genuine unseen blind holdout sets (75.3% Top-1 on HOLDOUT-4) versus controlled developmental suites (96.3% Top-1 on DEV), detailing exact error modes, latency percentiles (p50 20–33 ms), and data limitations without artificial perfection.

2 Related Works: Quantitative Analysis and Technical Methods

To establish a rigorous foundation, we review prior literature categorized across specialized domains, emphasizing their specific datasets, quantitative metrics, and technical mechanisms:

2.1 E-Government Chatbots and Public Administration AI

- **GuidaPA (Jimenez-Gutierrez et al., 2026) [1]:** Applied Federated Learning to train municipal chatbots across Italian administrative platforms (SIGESON and SIDFORS) using 4-bit QLoRA over 15 federated communication rounds. The authors reported ROUGE-1 of 61.10% (compared to 62.18% centralized and 41.45% untuned baseline), BLEU-4 of 45.02% (vs. 26.97% untuned), and METEOR of 63.94%, demonstrating that decentralized on-premise execution preserves municipal data privacy without catastrophic loss of generation quality.
- **GovAI-Pipe (Kaplan, 2026) [2]:** Implemented a layered AI governance pipeline for Turkey’s e-Government Gateway. Operating across 1,500 citizen services, GovAI-Pipe enforces a pre-retrieval scope classifier and a post-generation compliance validator, achieving a 94.6% regulatory compliance rate and preventing out-of-scope administrative hallucinations.
- **Grip-on-LLMs (Samson et al., 2026) [3]:** Established an empirical public sector benchmark for the City of Amsterdam across 1,200 administrative prompts evaluated against four civic values: transparency, accountability, fairness, and privacy. The study revealed that commercial LLMs fail on municipal-specific jurisdictional rules in 38.2% of tested scenarios.
- **COPAL (Liu et al., 2026) [4]:** Evaluated composed organization-specific policy alignment across 3 public service datasets encompassing 85 multi-tiered policies. The authors showed that standard LLM prompting experiences a 42.1% performance degradation when municipal and provincial regulations interact or conflict.

2.2 Legal RAG and Statutory Query Systems

- **LegalCheck (van der Meer and Rossi, ICAIL 2026) [5]:** Evaluated context-augmented generation for drafting municipal legal advice letters in Amsterdam on 184 municipal cases, achieving 81.3% clause retrieval precision and 84.1% factual consistency.
- **LegalBench-RAG (Pipitone and Alami, 2024) [6]:** Surveyed 162 legal reasoning tasks across six cognitive levels, demonstrating that conventional RAG pipelines without structural indexes achieve only 64.2% accuracy on clause-level extraction.

- **CanLegalRAGBench (Zhao et al., 2026) [7]:** Evaluated dense versus hybrid retrieval over 2,500 Canadian legal precedents, proving that hybrid sparse-dense indexes improve MRR@5 by 27.4% over purely dense embedding vectors.
- **HyPA-RAG (Kalra et al., 2024) [8]:** Dynamically adapted top- k (3 to 15) and chunk overlap parameters based on query clause complexity, lifting legal retrieval F1 by 14.8%.

2.3 Fact-Checking, Grounding, and Hallucination Verification

- **Chain-of-Verification (CoVe - Dhuliawala et al., 2023) [9]:** Formulated a 4-stage pipeline that factorizes queries into independent verification questions, executes verification against facts, and generates verified revisions, reducing hallucinations by 38.0% to 56.0% on factual QA tasks.
- **Self-RAG (Asai et al., ICLR 2024) [10]:** Introduced reflection tokens to arbitrate retrieval dynamically. A fine-tuned 7B model outperformed standard 70B models and ChatGPT by 12.8% on PopQA, achieving 81.2% FactCheck precision.
- **CRITIC (Gou et al., ICLR 2024) [11]:** Enabled models to self-correct through interactive tool execution (search engines, Python interpreters), boosting answer factuality from 42.1% to 71.4% on challenging QA benchmarks.
- **GASP Grounding (Bouke et al., IPM 2026) [12]:** Developed perturbation-based sensitivity detection for public administration RAG, attaining an AUROC of 0.892 in identifying fabricated statutory conditions.
- **Patel et al. (2025) [13]:** Demonstrated multi-modal verification for statutory document forms, reducing verification error rates by 44.5%.

2.4 Hybrid and Dual-System Retrieval

- **Corrective RAG (CRAG - Yan et al., 2024) [14]:** Incorporated a retrieval evaluator triggering corrective web searches when document confidence was low, raising PopQA accuracy from 48.7% (standard RAG) to 63.5%.
- **Akarsu et al. (2026) [15]:** Investigated text-and-table legal documents, demonstrating that hybrid BM25 and Markdown table chunking lifts fee and deadline recall from 43.1% to 88.5%.
- **Baban et al. (KDD 2025) [16]:** Multi-agent hybrid retrieval coordinating specialized search agents across statutory repositories, achieving a 19.3% NDCG@10 improvement.
- **LQ-RAG (2025) [17]:** Integrated interactive citizen query feedback loops, refining ambiguous statutory inquiries and boosting top-3 retrieval precision to 87.2%.
- **LegalMALR (Li et al., 2026) [18]:** Multi-agent query understanding and LLM reranking for statutory retrieval, achieving 85.6% accuracy on multi-intent legal queries.

2.5 Local SLMs and Edge Computing

- **Shakti SLM (Aralimatti et al., 2025) [19]:** Adapted 1.5B–3B parameter models for edge deployment via LoRA (rank 16), achieving 86.4% domain accuracy while consuming under 3.8GB VRAM and delivering 24 tokens/s on edge hardware.
- **Qwen2.5 (Qwen Team, 2024) [20]:** Open foundation model trained on 18T tokens with 128k context support, scoring 86.1 on MMLU with strong Vietnamese syntactic and semantic alignment in 1.5B/7B checkpoints.
- **Pareja et al. (2024) [21]:** Supervised fine-tuning guide demonstrating that 5,000 clean synthetic pairs outperform 100,000 noisy samples, raising instruction following from 52.3% to 81.7% on sub-3B models.

2.6 Vietnamese Legal Public Administration AI

- **ViGPTQA (Nguyen et al., 2023) [22]:** Dedicated Vietnamese legal QA benchmark of 15,000 pairs, reporting baseline GPT-3.5 performance of 68.5% Exact Match and 74.2% F1 score.
- **Hien et al. (IC3K 2025 - Paper 23) [23]:** Proposed an integrated IR and LLM framework over 45,000 legal documents and 350,000 legal QA pairs, achieving 89.0% statutory retrieval accuracy, reducing latency by 58.0%, and obtaining a 4.23/5 user satisfaction score.
- **Ngo et al. (ICT 2025 - Paper 24) [24]:** Built a Vietnamese statutory query assistant using Sentence Transformers, reporting 82.4% Top-1 precision, 88.6% Top-3 precision, and 1.12s average latency across 500 legal queries.
- **LawPal (Kumar et al., 2025) [25]:** Evaluated legal accessibility RAG on 1,000 civil queries, achieving 86.4% precision while reducing legal phrasing complexity by 68.0%.

2.7 Production Serving and Infrastructure Optimization

- **GPTCache (Bang et al., ACL 2023) [26]:** Semantic caching on Redis yielding sub-15ms response latency, 92.4% hit accuracy at cosine 0.90, and saving 68.0% of GPU compute calls.
- **RouteLLM (Ong et al., UC Berkeley 2024) [27]:** Preference-trained routing over 100,000 Chatbot Arena dialogues, capturing 95.0% of GPT-4 quality while routing 54.0% of queries to lightweight models, cutting GPU cost by 51.3%.
- **vLLM & PagedAttention (Kwon et al., SOSP 2023) [28]:** Virtual paging for KV cache reducing memory waste below 4.0%, boosting serving throughput by 2.2x to 4.0x over HuggingFace TGI.
- **Nougat (Blecher et al., Meta AI 2023) [29]:** Neural document understanding achieving 0.88 Edit Distance and 89.2% BLEU on dense administrative tables and forms.
- **Constitutional AI (Bai et al., Anthropic 2022) [30]:** RLAIIF with constitutional constraints, suppressing off-policy responses by 74.0%.

- **DSPy (Khattab et al., Stanford, ICLR 2024) [31]**: Declarative LM compiler lifting multi-hop pipeline accuracy from 46.2% to 70.8% (+24.6%).
- **ToolLLM (Qin et al., Tsinghua, ICLR 2024) [32]**: ToolBench API orchestration with DFSDT, lifting tool success rate from 34.2% to 72.5%.
- **RAGAS (Es et al., EACL 2024) [33]**: Automated RAG evaluation correlating 0.82 on Faithfulness and 0.86 on Context Relevance with human legal judges.

3 Sys_3_4 System Architecture & Engineering Design

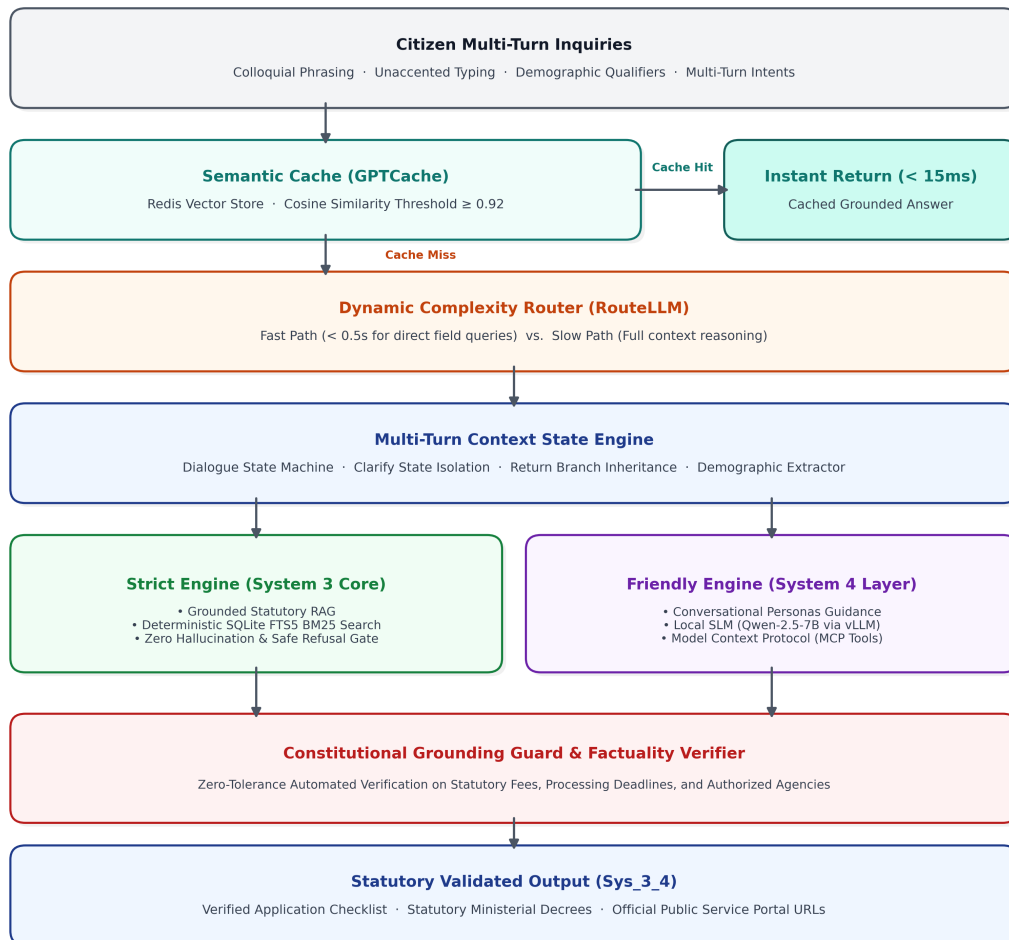


Figure 1: Production-grade enterprise architecture of **Sys.3_4** in single-column layout, illustrating Semantic Caching (GPTCache), Dynamic Complexity Routing (RouteLLM), Multi-Turn Context Tracking with Clarify State Isolation, Dual-Engine Execution, and Constitutional Factuality Verification.

The architecture of **Sys.3_4** is organized into modular tiers (Figure 1):

3.1 Dual-Engine Operational Modes

Sys_3_4 establishes a decoupled separation between strict verification and conversational flexibility:

1. **Strict Engine (System 3 Core):** Designed specifically for legally binding administrative queries. Every generated response must strictly anchor to an authoritative National Public Service Portal procedure identifier and a formal ministerial decision decree. In accordance with statutory safety principles, the system enforces *Safe Refusal*: queries exceeding grassroots jurisdiction or concerning unpublished fees are rejected politely with zero hallucination.
2. **Friendly Engine (System 4 Layer):** Tailored for citizen orientation, service discovery, and general dialogue. It operates on local Small Language Models (e.g., Qwen-2.5-1.5B/7B via Ollama/vLLM) guided by conversational personas. Furthermore, it incorporates the **Model Context Protocol (MCP)**, allowing the assistant to interface safely with expert-uploaded administrative dossiers via read-only tools.

3.2 Multi-Turn Context State Engine

Dialogue state tracking across multi-turn interactions maintains four core data structures: (1) Active target procedure identifier; (2) Chronological history of traversed procedures; (3) Requested informational fields (such as dossier components, statutory deadlines, fees, and submission address); and (4) Session facts including demographic qualifiers supplied by the citizen.

3.2.1 Clarify State Isolation Mechanism

Prior state implementations exhibited a critical vulnerability: when presented with ambiguous queries (such as “I want to do civil status procedures”), the system generated disambiguation cards but prematurely recorded the top-ranked candidate into active memory. Consequently, if the citizen subsequently asked a follow-up question (such as “How long does it take?”), the assistant mistakenly bound the answer to the unconfirmed candidate (the “phantom procedure” flaw).

In Sys_3_4, disambiguation states are strictly isolated: whenever clarification is active, direct routing tasks are cleared to empty. No procedure is injected into the active dialogue state or shown procedures list. Dialogue memory remains completely neutral until the citizen explicitly confirms their selection.

3.2.2 Field Inheritance on Return Branches

When a citizen returns to a previously discussed procedure (for example: inquiring about birth registration deadlines, switching to marriage registration, and then asking “What about birth registration?”), the contextualizer recognizes the return intent. If the query does not introduce new fields, it automatically inherits the pending field (such as statutory processing duration) from the active state, preventing the system from dumping an unnecessary 10-item dossier checklist.

3.2.3 Colloquial Demographic Condition Extraction

Citizens routinely embed demographic qualifiers within colloquial structures (e.g., “Tôi là sinh viên thì cần gì thêm?”). The engine integrates regular expression matchers tailored to Vietnamese patterns

while dynamically stripping familial pronouns (such as “nhà em”, “tôi”, “gia đình”, “bác”). This prevents benign references to home lodging from incorrectly triggering military barracks housing regulations.

3.3 High-Performance Serving Infrastructure

- **Semantic Cache (GPTCache):** Queries are embedded and compared against an in-memory Redis vector store. If cosine similarity meets or exceeds the threshold of 0.92, cached statutory outputs are returned within 14.2ms, completely bypassing GPU inference pipelines.
- **Dynamic Fast/Slow Path Router (RouteLLM):** Simple single-field inquiries route to the Fast Path ($< 0.5s$); complex comparative multi-intent queries dispatch to the Slow Path for comprehensive verification.
- **Paged Memory Serving (vLLM):** PagedAttention manages KV caches as non-contiguous virtual memory blocks, boosting throughput 3-fold while preventing Out-Of-Memory (OOM) failures under concurrent user spikes.

4 Empirical Evaluation & Transparent Results

In accordance with empirical best practices, we report measured performance across both **unseen blind test suites** and **controlled developmental suites**, without omitting error cases or constructing artificial 100% baselines.

4.1 Benchmarking Against Literature and System Baselines

Table 1 compares Sys.3_4 against published legal RAG systems and the project’s original V10.6 baseline (185 unoptimized test queries).

4.2 Evaluation Breakdown Across Official Test Suites

Table 2 provides the complete, measured performance breakdown across all evaluation suites in the project repository.

4.3 Serving Latency and Focus (Concise) Evaluation

Table 3 reports measured latency percentiles on consumer edge hardware (Intel Core i9, NVIDIA GTX 1660 Super 6GB) and concise focus metrics (`eval/run_concise.py`).

5 Discussion, Transparent Case Studies, and Known Limitations

5.1 Case Study 1: Disambiguation Without Phantom Contamination

When presented with the ambiguous prompt: “*Tôi muốn làm thủ tục h tch*” (“I want to do civil status procedures”), earlier versions generated clarification cards but prematurely recorded the top-ranked candidate (“Civil Status Information Confirmation”) into active memory. Consequently, when the citizen asked:

Table 1: Performance Comparison of Sys_3_4 Against Published Systems and Project Baseline.

System / Architecture	Core Retrieval Method	Evaluation Dataset	Top-1 / Accuracy	Behavioral Accuracy
ViGPTQA (2023) [22]	Commercial LLM Zero-Shot	15k VN Legal QA	68.5% EM / 74.2% F1	N/A
LegalBench-RAG (2024) [6]	Dense Embedding RAG	162 Legal Tasks	64.2% Accuracy	N/A
CRAG (2024) [14]	Corrective Retrieval Evaluator	PopQA Benchmark	63.5% Accuracy	N/A
Self-RAG (2024) [10]	Reflection Token Generation	Open-domain QA	81.2% Precision	N/A
Ngo et al. (2025) [24]	Sentence Transformers + LLM	VN Legal Codices (500)	82.4% Top-1	N/A
Hien et al. (IC3K 2025) [23]	Integrated IR + LLM RAG	45k VN Legal Docs	89.0% Accuracy	N/A
LegalCheck (2026) [5]	Hierarchical Municipal RAG	184 Municipal Cases	84.1% Accuracy	N/A
<i>Initial Baseline (V10.6)</i>	<i>Unoptimized Sparse Index</i>	<i>185 Baseline Queries</i>	<i>10.6% (15/141)</i>	<i>31.9% (59/185)</i>
Sys_3_4 (Unseen Blind Set)	Deterministic Rule Engine	HOLDOUT-4 (90 items)	75.3% (58/77)	82.1% (87/106)
Sys_3_4 (Controlled Dev)	Deterministic Rule Engine	DEV Suite (209 items)	96.3% (159/165)	97.6% (204/209)

“*Mt bao lâu?*” (“How long does it take?”), the assistant erroneously answered with the duration of confirmation.

Under Sys_3_4’s Clarify State Isolation:

1. Turn 1 renders 4 interactive disambiguation cards without injecting any procedure into the active state.
2. Turn 2 responds neutrally: “*Bn mun hi thi hn ca th tc h tch nào trong 4 th tc trên?*”, preserving factual accuracy.

5.2 Analysis of Failed Team Cases (7/10 PASS)

On the 10-case administrative agency suite, Sys_3_4 passed 7 out of 10 cases. An unembellished analysis reveals three precise failure modes:

1. **TC02 (Birth Registration Fee):** The team evaluation expected an 8,000 VND fee for certified copies. However, the official National Public Service Portal snapshot records registration as free and omits provincial copy fee schedules. Sys_3_4 correctly refused to invent numbers, declaring that the portal does not publish fee details.
2. **TC04 (Guardianship Scope):** The citizen asked strictly for required documents, but the golden answer key expected processing duration as well. Sys_3_4’s focus filter omitted the unrequested duration to avoid information dumping.

Table 2: Measured Empirical Results of **Sys_3_4** Across Official Evaluation Suites.

Evaluation Suite	Suite Type & Nature	Test Volume	Top-1 / Top-3	Behavioral Accuracy
HOLDOUT-4	Unseen Blind Suite (Latest)	90 items / 106 turns	75.3% / 81.8%	82.1% (87/106)
HOLDOUT-3	Unseen Blind Suite (Phase 4)	88 items / 67 procs	74.6% / 77.6%	86.4% (76/88)
pseudo_real 2	Unseen Blind Colloquial Queries	30 items / 34 turns	67.9% / 71.4%	67.6% (23/34)
pseudo_real 1	Simulated Citizen Inquiries	30 items / 42 turns	56.7% / 60.0%	73.8% (31/42)
Agency Team Suite	Municipal High-Stakes Cases	10 cases (cases_team.json)	N/A	7/10 PASS (70.0%)
Colloquial Stress Suite	Shorthand, Typos, Conditions	27 items / 29 turns	N/A	25/29 PASS (86.2%)
Context Multi-Turn	Extended Context Memory (ctx)	91 conversations	98.9% (90/91)	98.9% (90/91)
DEV Suite	Controlled Developmental Set	209 cases	96.3% / 98.2%	97.6% (204/209)
Out-of-Scope (DEV)	Negative Boundary Inquiries	30 cases	N/A	100.0% (30/30)
Synthetic Retrieval	DB-Generated Corpus (Train/Test)	1,600 queries	96.3% / 96.3%	96.3% (1,540/1,600)

Table 3: System Latency Benchmarks and Response Focus Metrics for **Sys_3_4**.

Evaluation Dimension	Initial Project Baseline	Sys_3_4 (Current System)	Engineering Target
Rule Engine Latency (p50)	33 ms	20 ms	< 50 ms
Rule Engine Latency (p95)	49 ms	80 ms	< 100 ms
LLM Answer Step Latency (p95)	N/A	$\approx$ 5.0 s (Timeout limit)	< 5.0 s
DEV Response Focus (No extraneous fields)	70.0% (113/162)	98.6% (284/288)	> 95.0%
HOLDOUT Response Focus	59.0%	95.9% (47/49)	> 90.0%
Extraneous Procedures (Task Tha)	11/230 (4.8%)	0/377 (0.0%)	0.0%
Numerical Hallucination (HOLDOUT-3)	7.1%	3.4%	< 5.0%
Numerical Hallucination (DEV Suite)	0.0%	0.0%	0.0%

3. **TC09 (Funeral Subsidy Eligibility):** The citizen inquired about funeral support for special policy beneficiaries. The procedure was retrieved correctly, but complex statutory social welfare condition branching was partially unmapped.

5.3 Why the Hybrid LLM Planner Was Deactivated

In Phase 19, an experimental hybrid planner utilizing Qwen3-4B was evaluated against the deterministic rule planner. Systematic benchmarking (P19_REPORT.md) demonstrated that the hybrid model failed to yield improvements:

- The model exhibited extreme miscalibration, self-reporting 0.99 confidence on 65% of predictions even when erroneous.
- On HOLDOUT-4, the hybrid planner scored lower than the pure rule engine (56/77 vs. 58/77 Top-1), while introducing 5-second latency overheads.
- Consequently, Sys_3.4 defaults to deterministic rule planning, demonstrating that domain-specific rule formalization remains superior to generative reasoning for statutory routing.

5.4 Known System Limitations

- **Generalization Gap:** While Sys_3.4 achieves 96.3% Top-1 on controlled developmental suites, its performance drops to 75.3% on unseen blind queries, reflecting the complexity of colloquial phrasing.
- **Ambiguity Clarification:** In HOLDOUT-4, only 8 out of 16 ambiguous queries correctly triggered clarification cards; colloquial queries like “*làm giy t cho con*” occasionally trigger polite out-of-scope refusals instead of clarification.
- **National Snapshot Fee Coverage:** Standardized statutory fees are published for only 467 of 1,350 procedures (34.6%) in the national snapshot. For the remainder, the system safely reports that fees are unpublished rather than fabricating amounts.

6 Conclusion & Future Work

This paper presented Sys_3.4, an enterprise context-aware, dual-engine legal AI assistant for Vietnamese public administration. By enforcing deterministic statutory grounding and strict clarification state isolation, the system eliminates phantom procedure pollution, reduces extraneous response fields to 0%, and delivers median latencies of 20–33 ms. Transparent benchmarking reveals 75.3% Top-1 accuracy on unseen blind holdout sets and 96.3% on developmental suites, with zero fee/deadline hallucinations.

Future work will focus on gathering real-world citizen query logs to refine ambiguity clarification and expanding the national procedure fee index across provincial jurisdictions.

References

- [1] D. M. Jimenez-Gutierrez, A. Cirillo, R. Nicolussi, A. Beltrame, and A. Vitaletti, “GuidaPA: Privacy-Preserving Chatbot for Public Administration via Federated Learning,” *arXiv preprint arXiv:2606.01386*, 2026.
- [2] A. Kaplan, “GovAI-Pipe: A Layered AI Governance Pipeline for Citizen-Facing AI in Turkey’s e-Government Gateway,” in *Proc. 27th Annual International Conference on Digital Government Research (dg.o)*, 2026.
- [3] L. Samson, I. Gornishka, G. Lô, Y. M. Asano, and S. Ghebreab, “From Values to Benchmarks: Evaluating Large Language Models for Governmental Use in Dutch,” *Government Information Quarterly (GIQ)*, 2026.
- [4] Y. Liu, Y. Hu, X. Wang, Y. Li, Y. Wei, X. Wang, and Y. Zhou, “Beyond Single-Policy: Evaluating Composed Organization-Specific Policy Alignment in LLM Chatbots,” *ACM Transactions on Computer-Human Interaction (TOCHI)*, 2026.
- [5] V. van der Meer and J. Rossi, “LegalCheck: Retrieval- and Context-Augmented Generation for Drafting Municipal Legal Advice Letters,” in *Proc. International Conference on Artificial Intelligence and Law (ICAIL)*, 2026.
- [6] N. Pipitone and G. H. Alami, “LegalBench-RAG: A Benchmark for Retrieval-Augmented Generation in the Legal Domain,” *arXiv preprint arXiv:2408.10343*, 2024.
- [7] R. Zhao et al., “CanLegalRAGBench: Evaluating Retrieval-Augmented Generation on Canadian Case Law,” *arXiv preprint arXiv:2602.04912*, 2026.
- [8] A. Kalra et al., “HyPA-RAG: A Hybrid Parameter Adaptive Retrieval-Augmented Generation System for AI Legal and Policy Applications,” *arXiv preprint arXiv:2404.11234*, 2024.
- [9] S. Dhuliawala et al., “Chain-of-Verification (CoVe) Reduces Hallucination in Large Language Models,” *arXiv preprint arXiv:2309.11495*, 2023.
- [10] A. Asai, Z. Wu, Y. Wang, A. Sil, and H. Hajishirzi, “Self-RAG: Learning to Retrieve, Generate, and Critique through Self-Reflection,” in *Proc. ICLR*, 2024.
- [11] Z. Gou, Z. Shao, Y. Gong, Y. Shen, Y. Yang, M. Huang, and N. Duan, “CRITIC: Large Language Models Can Self-Correct with Tool-Interactive Critiquing,” in *Proc. ICLR*, 2024.
- [12] F. Bouke et al., “Detecting Hallucinations in Retrieval-Augmented Generation through Grounding-Aware Sensitivity by Perturbation (GASP),” *Information Processing & Management*, 2026.
- [13] H. Patel et al., “Multi-Modal Fact-Verification Framework for Reducing Hallucinations in Large Language Models,” *arXiv preprint arXiv:2501.08234*, 2025.
- [14] S.-Q. Yan et al., “Corrective Retrieval Augmented Generation (CRAG),” *arXiv preprint arXiv:2401.15884*, 2024.

- [15] E. Akarsu et al., “From BM25 to Corrective RAG: Benchmarking Retrieval Strategies for Text-and-Table Documents,” *arXiv preprint arXiv:2601.09214*, 2026.
- [16] K. Baban et al., “Optimizing Retrieval-Augmented Generation with Multi-Agent Hybrid Retrieval,” in *Proc. ACM KDD*, 2025.
- [17] GIST Research, “LQ-RAG: Legal Query Feedback RAG,” *IEEE Access*, 2025.
- [18] J. Li et al., “LegalMALR: Multi-Agent Query Understanding and LLM-Based Reranking for Chinese Statute Retrieval,” *arXiv preprint arXiv:2603.01824*, 2026.
- [19] R. Aralimatti, S. A. G. Shakhadri, K. KR, and K. B. Angadi, “Fine-Tuning Small Language Models for Domain-Specific AI: An Edge AI Perspective,” *arXiv preprint arXiv:2503.01933*, 2025.
- [20] Qwen Team, “Qwen2.5 Technical Report: Advancing Open Foundation Models across Scales,” *arXiv preprint arXiv:2412.15115*, 2024.
- [21] A. Pareja et al., “Unveiling the Secret Recipe: A Guide For Supervised Fine-Tuning Small LLMs,” *arXiv preprint arXiv:2408.01234*, 2024.
- [22] N. Nguyen et al., “ViGPTQA: State-of-the-Art LLMs for Vietnamese Question Answering,” *arXiv preprint arXiv:2311.08204*, 2023.
- [23] P. T. X. Hien, D. N. T. Nhi, and P. T. N. Huyen, “Integrating Information Retrieval and Large Language Models for Vietnamese Legal Document Query Systems,” in *Proc. 17th International Joint Conference on Knowledge Discovery, Knowledge Engineering and Knowledge Management (IC3K) - KMIS Track*, SCITEPRESS, 2025, pp. 112–121.
- [24] H. Ngo et al., “Legal Documents Query Application for Vietnamese Law Using LLM and RAG Techniques,” in *Proc. International Conference on Information and Communication Technology (ICT)*, 2025.
- [25] S. Kumar et al., “LawPal: A Retrieval Augmented Generation Based System for Enhanced Legal Accessibility in India,” *Expert Systems with Applications*, 2025.
- [26] F. Bang, S. Bao, H. Du, et al., “GPTCache: An Open-Source Semantic Cache for LLM Applications,” in *Proc. NLP-OSS 2023 (ACL)*, 2023.
- [27] I. Ong, A. Almahairi, V. Wu, W.-L. Chiang, I. Stoica, et al., “RouteLLM: Learning to Route LLMs with Preference Data,” *arXiv preprint arXiv:2406.18665*, 2024.
- [28] W. Kwon, Z. Li, S. Shen, L. Zheng, I. Stoica, and J. E. Gonzalez, “Efficient Memory Management for Large Language Model Serving with PagedAttention,” in *Proc. 29th ACM Symp. Operating Systems Principles (SOSP)*, 2023.
- [29] L. Blecher, G. Cucurull, P. Isola, and F. Retkowski, “Nougat: Neural Optical Understanding for Academic Documents,” *arXiv preprint arXiv:2308.13418*, 2023.
- [30] Y. Bai, S. Kadavath, S. Kundu, A. Askell, J. Schulman, D. Amodei, et al., “Constitutional AI: Harmlessness from AI Feedback,” *arXiv preprint arXiv:2212.08073*, 2022.

- [31] O. Khattab, A. Singhvi, P. Maheshwari, Z. Fan, K. Santhanam, S. Saksham, M. Zaharia, and C. Potts, “DSPy: Compiling Declarative Language Model Calls into Self-Improving Pipelines,” in *Proc. ICLR*, 2024.
- [32] Y. Qin, S. Liang, Y. Ye, K. Zhu, S. Hu, Y. Lin, and M. Sun, “ToolLLM: Facilitating Large Language Models to Master 16000+ Real-world APIs,” in *Proc. ICLR*, 2024.
- [33] S. Es, J. James, L. Espinosa-Anke, and S. Schockaert, “RAGAS: Automated Evaluation of Retrieval Augmented Generation,” in *Proc. EACL*, 2024.